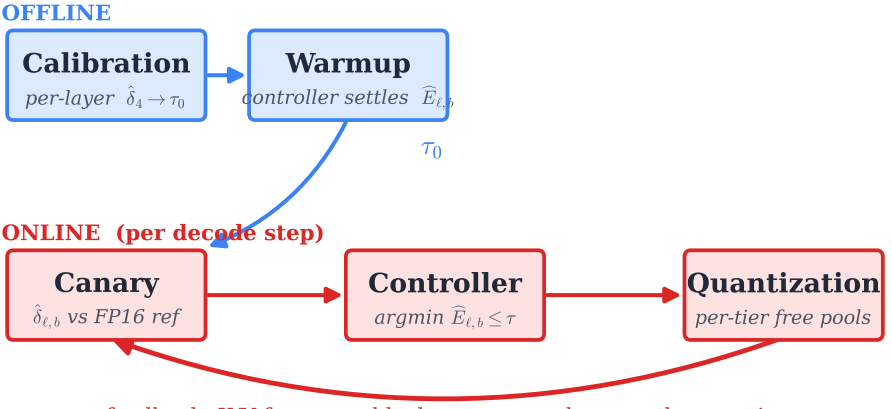
OFFLINE
Calibration
per-layer $\hat{\delta}_4 \to \tau_0$
Warmup
controller settles $\widehat{E}_{\ell,b}$
τ_0
ONLINE (per decode step)
Canary
$\hat{\delta}_{\ell,b}$ vs FP16 ref
Controller
argmin $\widehat{E}_{\ell,b} \le \tau$
Quantization
per-tier free pools
feedback: K/V from new blocks → canary observes the new tier